



SIMATS Online Programming Lab

JAVA PROGRAMMING

 SANGEPU VENKATA PRAVEEN
192373013RD

Palindrome

PALINDROME

Write a Java program to check if a given string is a palindrome or not.

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String s = sc.nextLine();
6         s = s.replaceAll("\\s+", "").toLowerCase();
7         int i = 0, j = s.length() - 1;
8         boolean isPalindrome = true;
9         while (i < j) {
10             if (s.charAt(i) != s.charAt(j)) {
11                 isPalindrome = false;
12                 break;
13             }
14             i++;
15             j--;
16         }
17         System.out.print(isPalindrome ? "Palindrome" : "Not Palindrome");
18         sc.close();
19     }
20 }
```

OUTPUT

Palindrome

racecar



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Occurrence

OCCURRENCE

Write a Java program to count the number of occurrences of a given character in a string.

```
1 import java.util.Scanner;
2 public class Occurrence {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         char target = sc.next().charAt(0);
7         int count = 0;
8         for (int i = 0; i < str.length(); i++) {
9             if (str.charAt(i) == target) {
10                 count++;
11             }
12         }
13         System.out.println(count);
14         sc.close();
15     }
16 }
17 }
```

OUTPUT

2

programming
m



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Check

CHECK

Write a Java program to check if a given string contains only digits.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String input = sc.nextLine();
6         if (isOnlyDigits(input)) {
7             System.out.println("True");
8         } else {
9             System.out.println("False");
10        }
11    }
12    public static boolean isOnlyDigits(String str) {
13        if (str == null || str.isEmpty()) {
14            return false;
15        }
16        for (char c : str.toCharArray()) {
17            if (!Character.isDigit(c)) {
18                return false;
19            }
20        }
21        return true;
22    }
23 }
```

OUTPUT

True

12345



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Length

LENGTH

Write a Java program to find the length of the longest substring without repeating characters in a given string.

```
1 import java.util.*;
2 public class LongestSubstring {
3     public static int lengthOfLongestSubstring(String s) {
4         int n = s.length();
5         int maxLength = 0;
6         Map<Character, Integer> map = new HashMap<>();
7         int left = 0;
8         for (int right = 0; right < n; right++) {
9             if (map.containsKey(s.charAt(right))) {
10                left = Math.max(map.get(s.charAt(right)) + 1, left);
11            }
12            map.put(s.charAt(right), right);
13            maxLength = Math.max(maxLength, right - left + 1);
14        }
15        return maxLength;
16    }
17    public static void main(String[] args) {
18        Scanner sc = new Scanner(System.in);
19        if (sc.hasNextLine()) {
20            String input = sc.nextLine();
21            System.out.println(lengthOfLongestSubstring(input));
22        }
23    }
24 }
```

OUTPUT

3

abcbcbbb



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Lower

LOWER

Write a Java program to convert a given string to lowercase.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         if (sc.hasNextLine()) {
6             String input = sc.nextLine();
7             System.out.println(input.toLowerCase());
8         }
9         sc.close();
10    }
11 }
```

OUTPUT

hello world!

Hello World!



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Remove

REMOVE

Write a Java program to remove all white spaces from a given string.

```
1 public class RemoveWhitespace {
2     public static void main(String[] args) {
3         String input = "Hello World from SIMATS";
4         String result = input.replaceAll("\\s", "");
5         System.out.println("Original: " + input);
6         System.out.println("Modified: " + result);
7     }
8 }
```

OUTPUT

Original: Hello World from SIMATS
Modified: HelloWorldfromSIMATS

Your input goes here!



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Reverse

REVERSE

Write a Java program to reverse a given string.

```
1 import java.util.Scanner;
2 public class ReverseString {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String input = sc.nextLine();
6         String reversed = new StringBuilder(input).reverse().toString();
7         System.out.println(reversed);
8     }
9 }
```

OUTPUT

olleH

Hello



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Concatenate

CONCATINATE

Write a Java program to concatenate two given strings.

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         String s1 = sc.nextLine();
8         String s2 = sc.nextLine();
9
10        System.out.print(s1 + s2);
11
12        sc.close();
13    }
14 }
```

OUTPUT

HelloWorld

Hello
World



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Repeat

REPEAT

Write a Java program to find the first non-repeated character in a given string.

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String s = sc.nextLine();
6         int[] freq = new int[256];
7         for (int i = 0; i < s.length(); i++) {
8             freq[s.charAt(i)]++;
9         }
10        for (int i = 0; i < s.length(); i++) {
11            if (freq[s.charAt(i)] == 1) {
12                System.out.println(s.charAt(i));
13                return;
14            }
15        }
16        System.out.println("-1");
17    }
18 }
```

OUTPUT

programming



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Frequent

FREQUENT

Write a Java program to find the most frequent word in a given string.

```
1 import java.util.*;
2 public class FrequentWord {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String input = sc.nextLine();
6         String[] words = input.toLowerCase().split("\\s+");
7         Map<String, Integer> wordCount = new HashMap<>();
8         for (String word : words) {
9             word = word.replaceAll("[^a-zA-Z]", "");
10            if (!word.isEmpty()) {
11                wordCount.put(word, wordCount.getOrDefault(word, 0) + 1);
12            }
13        }
14        String mostFrequent = "";
15        int maxCount = 0;
16        for (Map.Entry<String, Integer> entry : wordCount.entrySet()) {
17            if (entry.getValue() > maxCount) {
18                maxCount = entry.getValue();
19                mostFrequent = entry.getKey();
20            }
21        }
22    }
23 }
```

OUTPUT

a

This is a test this is only a test

Substring

SUBSTRING

Write a Java program to create a substring from a given string.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         int start = sc.nextInt();
7         int end = sc.nextInt();
8         String sub = str.substring(start, end);
9         System.out.println(sub);
10
11         sc.close();
12     }
13 }
```

abcdef
1
4

OUTPUT

bcd

Compare

COMPARE

Write a Java program to compare two strings lexicographically.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String s1 = sc.nextLine();
6         String s2 = sc.nextLine();
7         int result = s1.compareTo(s2);
8         if (result < 0) {
9             System.out.println("First string is lexicographically smaller.");
10        } else if (result > 0) {
11            System.out.println("First string is lexicographically greater.");
12        } else {
13            System.out.println("Both strings are lexicographically equal.");
14        }
15        sc.close();
16    }
17 }
```

Hello
Hello

OUTPUT

Both strings are lexicographically equal.



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Suffix

SUFFIX

Write a Java program to check if a given string ends with a specific suffix.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         String suffix = sc.nextLine();
7         if (str.endsWith(suffix)) {
8             System.out.println("True");
9         } else {
10            System.out.println("False");
11        }
12
13        sc.close();
14    }
15 }
```

OUTPUT

True

HelloWorld
World



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Find

FIND

Write a Java program to find the index of a given character or substring in a given string.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         String target = sc.nextLine();
7         int index = str.indexOf(target);
8         System.out.println(index);
9         sc.close();
10    }
11 }
```

OUTPUT

6

hello world
world



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Upper

UPPER

Write a Java program to convert a given string to uppercase.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         if (sc.hasNextLine()) {
6             String input = sc.nextLine();
7             System.out.println(input.toUpperCase());
8         }
9         sc.close();
10    }
11 }
```

HELLO WORLD

Hello World

PC innv

Search

ENG IN



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RM-Char

RM-CHAR

Write a Java program to remove a specified character from a given string.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         char charToRemove = sc.next().charAt(0);
7         String result = str.replace(String.valueOf(charToRemove), "");
8         System.out.println(result);
9         sc.close();
10    }
11 }
```

HELL WRLD

Hello World
o

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Replace

REPLACE
Write a Java program to replace a specified character with another character in a given string.

```
1 import java.util.Scanner;
2 public class ReplaceChar {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String str = sc.nextLine();
6         char oldChar = sc.next().charAt(0);
7         char newChar = sc.next().charAt(0);
8         String result = str.replace(oldChar, newChar);
9         System.out.println(result);
10
11     }
12 }
13 }
```

hello world
o
a

OUTPUT
hella world

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Split

SPLIT
Write a Java program to split a given string into an array of substrings based on a specified delimiter.

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         String str = "apple,banana,cherry";
5         String delimiter = ",";
6         String[] substrings = str.split(java.util.regex.Pattern.quote(delimiter));
7         System.out.println(Arrays.toString(substrings));
8     }
9 }
```

apple,banana,cherry

OUTPUT
[apple, banana, cherry]

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Trim

TRIM

Write a Java program to trim a given string and remove leading and trailing spaces.

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String input = sc.nextLine();
6         System.out.println(input.trim());
7     }
8 }
```

OUTPUT

Hello World

Hello World

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Convert

CONVERT

Write a Java program to convert a given string to a char array.

```
1 public class Main {
2     public static void main(String[] args) {
3         String str = "Hello";
4         char[] charArray = str.toCharArray();
5         for (char c : charArray) {
6             System.out.print(c + " ");
7         }
8     }
9 }
```

OUTPUT

Hello

Hello